Type MS · Explain how a blocked coronary artery damages the heart

Answer 1 of 8 · Question 7 part a · 4 marks

2024 JUN · Paper 02 · find part a in this mark scheme extract

7 (a)	Any four from: • less blood flow/blood flow slows/restricted; • to heart muscle/tissue/cells; • less oxygen/glucose; • less aerobic respiration; • heart muscle/tissue/cells die/angina/heart attack;		(4)
(b) (i)	• (women 29 ÷ 100 (x 33 500 000) 9 715 000; • (men 26 ÷ 100 (x 33 000 000) 8 580 000; • (9 715 000 – 8 580 000) 1 135 000/1.135 million/1.135×10^6;	ecf for mp3	(3)
(ii)	• calculate BMI; • mass÷(height)^2; • compare result to BMI chart/ obese if 30 or more;	Accept • waist: hip ratio; • waist÷hip • Compare result to hip:waist ratio chart/	(3)
(c)	• beta blockers bind/ attach to receptors (on cardiac muscle); • prevent neurotransmitters from binding/attaching (to receptors); • reduce heart rate/reduced blood pressure;	Accept beta blockers fits into receptors accept neurotransmitters do not fit into receptors	(3)

Total for Question 7 = 13 marks

Type MS · Explain how a blocked coronary artery damages the heart

Answer 2 of 8 · Question 9 part a.ii · 3 marks

2019 JUN · Paper 02 · extract 1 of 2 · find part a.ii in this mark scheme extract

9 (a) (i)	• reference to platelets; • fibrinogen converted to fibrin; • (fibrin) mesh/crosslinks/fibres formed;		3
(ii)	• reduced blood flow to heart/cardiac muscle / tissue/cells; • less oxygen/glucose delivered; • less (aerobic) respiration/energy released (by cardiac tissue);		3
(b)(i)	size of blood clot/volume of water /aspirin solution used/temperature/shape of blood clot/width of straw;	Allow valid alternatives	1
(ii)	use water without aspirin dissolved;		1
(iii)	• the more tablets dissolved the more solution collected; • the more tablets dissolved the quicker the solution is collected; • aspirin dissolves/acts on blood clot; • size of blood clot reduced more quickly with more tablets/blood clot dissolves/breaks down faster with more tablets;		Max 3
(c)	Any four from: • aspirin binds to enzyme/aspirin competes (with substrate) for active site; • active site of enzyme changed/blocked; • less/no substrate binds (to enzyme); • reaction involving release of chemicals reduced/stopped; • platelets do not stick together;		Max 4

Total question 9 = 15 marks

Type MS · Explain how a blocked coronary artery damages the heart

Answer 2 of 8 · Question 9 part a.ii · 3 marks

2019 JUN · Paper 02 · extract 2 of 2 · find part a.ii in this mark scheme extract

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Type MS · Explain how a blocked coronary artery damages the heart

Answer 3 of 8 · Question 3 part b.i · 1 mark

2020 NOV · Paper 02 (regional R) · extract 1 of 2 · find part b.i in this mark scheme extract

Number				
3 (a)	(i)	A (1) (aorta) B it is not the pulmonary artery C it is not the pulmonary vein D it is not the vena cava		1
	(ii)	C (1) (right atrium) A it is not connected to the left atrium B it is not connected to the left ventricle D it is not connected to the right ventricle		1
(b)	(i)	circle drawn beyond by-pass vessel and around end of coronary vessels (1)		1
	(ii)	• no/less blood to heart muscles (1) • no/less oxygen/glucose transported (1) • no/less CO_2 removed (1)		3
	(iii)	• takes blood beyond blockage (1) • to supply heart muscles (1)		2

Total for question = 8 marks

Type MS · Explain how a blocked coronary artery damages the heart

Answer 3 of 8 · Question 3 part b.i · 1 mark

2020 NOV · Paper 02 (regional R) · extract 2 of 2 · find part b.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain how a blocked coronary artery damages the heart

Answer 4 of 8 · Question 3 part b.ii · 3 marks

2020 NOV · Paper 02 (regional R) · extract 1 of 2 · find part b.ii in this mark scheme extract

Number				
3 (a)	(i)	A (1) (aorta) B it is not the pulmonary artery C it is not the pulmonary vein D it is not the vena cava		1
	(ii)	C (1) (right atrium) A it is not connected to the left atrium B it is not connected to the left ventricle D it is not connected to the right ventricle		1
(b)	(i)	circle drawn beyond by-pass vessel and around end of coronary vessels (1)		1
	(ii)	• no/less blood to heart muscles (1) • no/less oxygen/glucose transported (1) • no/less CO_2 removed (1)		3
	(iii)	• takes blood beyond blockage (1) • to supply heart muscles (1)		2

Total for question = 8 marks

Type MS · Explain how a blocked coronary artery damages the heart

Answer 4 of 8 · Question 3 part b.ii · 3 marks

2020 NOV · Paper 02 (regional R) · extract 2 of 2 · find part b.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain how a blocked coronary artery damages the heart

Answer 5 of 8 · Question 7 part c · 3 marks

2023 NOV · Paper 01 · extract 1 of 2 · find part c in this mark scheme extract

7 (a)	pulmonary artery;		1
(b)	• Y is thicker; • blood pumped into aorta; • takes/distributes blood to organs/tissues/whole body; • requires high pressure • thicker wall means more muscle for stronger contraction;	ORA for each point	5
(c)	any three of • heart muscle deprived of oxygen; • insufficient energy for muscle contraction/less aerobic respiration/(more) anaerobic respiration; • heart stops pumping blood/pumps less blood; • heart attack;		3
(d)	• transfers CO_2 (from muscle cells) to blood capillary; • transfers oxygen from blood/capillary (to muscle cells); • transfers glucose from blood/capillary (to muscle cells);		3

Total 12 marks

Type MS · Explain how a blocked coronary artery damages the heart

Answer 5 of 8 · Question 7 part c · 3 marks

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Type MS · Explain how a blocked coronary artery damages the heart

Answer 6 of 8 · Question 6 part a · 4 marks

2024 JUN · Paper 02 (regional R) · find part a in this mark scheme extract

6	(a)	• reduced blood flow to the heart/cardiac muscle/coronary arteries; • less oxygen to heart/cardiac muscle/tissue; • less (aerobic) respiration/less energy for contraction; • (cardiac) muscle/cells die; • (leading to) heart attack/heart disease;		(4)
	(b) (i)	As the use of statins increases the level of blood cholesterol decreases;	allow inversely proportional	(1)
	(ii)	1988 = 31; 2005 = 17; 1 : 1.8/1 : 0.54/1:0.548/1:0.55;	ecf for incorrect graph readings. if use 18:17 ratio is 1:1.0588 (accept 1:1.059, 1.06, 1.1) full marks for correct final answer	(3)
	(iii)	• select sample of people; • record the number of people taking statins/not taking statins; • take blood samples; • measure the concentration/level of cholesterol (in blood); • repeat every year/between 1988-2008;		(5)
	(iv)	any two from the following: • gender/age not taken into account; • sample size maybe too small; • reference to a named lifestyle not taken into account e.g. diet, level of activity/smoking; • genetic predisposition/diseases not taken into account;		(2)

Total for Question 6 = 15 marks

Type MS · Explain how a blocked coronary artery damages the heart

Answer 7 of 8 · Question 4 part c · 2 marks

2024 NOV · Paper 01 · extract 1 of 2 · find part c in this mark scheme extract

4 (a)	(i)	• glucose (units); • linked/joined together;		2
	(ii)	insulin;		1
	(iii)	glucagon:		1
(b)	(i)	• $67.0 \div 1.69^2$; • 23.5/23.4/23.46/23.458;		2
	(ii)	healthy weight *less than 18.5 underweight* *18.5 to 24.9 healthy weight* *25 to 29.9 overweight* *30 to 34.9 obese* *35 to 39.9 severely obese* *more than 39.9 morbidly obese*	ECF value in (b)(i)	1
(c)		Any two from: • fatty deposits/plaques/atheroma; • in coronary arteries; • reduced blood flow to heart muscles/tissue; • lack of oxygen/glucose to heart muscle/tissue;		2

Total Marks 9

Type MS · Explain how a blocked coronary artery damages the heart

Answer 7 of 8 · Question 4 part c · 2 marks

2024 NOV · Paper 01 · extract 2 of 2 · find part c in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain how a blocked coronary artery damages the heart

Answer 8 of 8 · Question 4 part d · 3 marks

2021 JUN · Paper 02 · extract 1 of 2 · find part d in this mark scheme extract

Question number					
4	(a)	(i)	• eat too much; • too little exercise; • increased fat storage;		3
		(ii)	• heart disease/attacks/strokes; • causes joints in legs to develop arthritis;	Allow high blood pressure/hypertension	2
	(b)	(i)	C (pancreas); A is for excretion B is site of insulin action D produces other hormones		1
		(ii)	B (liver); A is for excretion C is site of production D produces other hormones		1
	(c)	(i)	26378 - 22092 = 4286; $\frac{4286 \times 100}{22092}$; = 19.4%;	Allow $\frac{26378 - 22092}{22092}$ for 2 marks Ecf from first calculation max 2 marks	3
		(ii)	0.9 x 3.8 = 3.4(2) million;		1
	(d)		any three from • less blood flows; • death of tissues; • increased risk of infection; • less white blood cells to deal with infection;		3

Total 14 marks

Type MS · Explain how a blocked coronary artery damages the heart

Answer 8 of 8 · Question 4 part d · 3 marks

2021 JUN · Paper 02 · extract 2 of 2 · find part d in this mark scheme extract

Question number	Answer	Notes	Marks